POLITECHNIKA ŚWIĘTOKRZYSKA
KIELCE UNIVERSITY OF TECHNOLOGY

Faculty of Electrical Engineering, Automatic Control and Computer Science

Study course: Computer Science
Course: Advanced database techniques
Group 11-B
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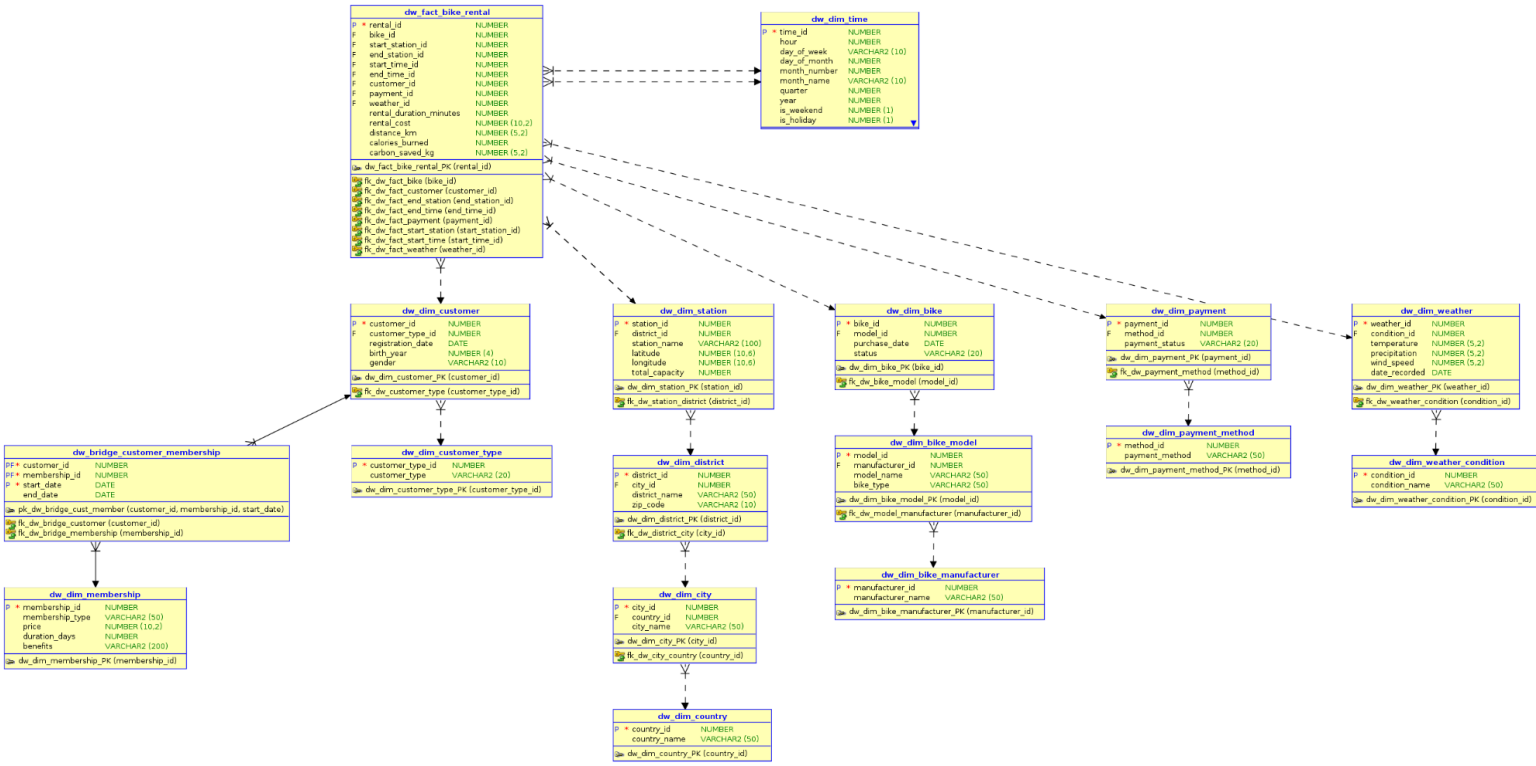
Report

1. Main structure

Subject of our project was to create a warehouse that will be able to store detailed information about bike rents. Main tables of the warehouse are: ‘customer’, ‘station’, ‘bike’, ‘payment’, ‘weather’ and main fact table ‘bike_rental’.



Structure of warehouse

The SQL script that generates this structure is located in file ‘ProjectADBT.sql’. Every table has a few constraints such as foreign key, primary key. There are also a few ‘ctl’ and ‘csv’ files which are used for filling our warehouse with data via sql loader. File that fills the warehouse is named ‘load_all.bat’. There are two Python files, which store the code for generating the csv files.

2. Queries

There are 5 different categories of analytical queries that we were supposed to implement (ROLLUP, CUBE, PARTITION, VIEWS, RANKING FUNCS). All these queries are located in file ‘queries.sql’, for each category we have written 3 different queries with comments above them. Here are a few of them:

```
-- 1. Top 5 cities by total rental revenue
SELECT *
FROM (
   SELECT
      ci.city_name,
      co.country_name,
      SUM(f.rental_cost) AS total_revenue,
      RANK() OVER (ORDER BY SUM(f.rental_cost) DESC) AS
city_rank
   FROM dw_fact_bike_rental f
   JOIN dw_dim_station s ON f.start_station_id = s.station_id
   JOIN dw_dim_district d ON s.district_id = d.district_id
   JOIN dw_dim_city ci ON d.city_id = ci.city_id
   JOIN dw_dim_country co ON ci.country_id = co.country_id
   GROUP BY ci.city_name, co.country_name
) WHERE city_rank <= 5;
```

```
-- 1. View: City-Daily Rental Stats
        CREATE OR REPLACE VIEW vw_city_day_rentals AS
        SELECT
            ci.city_name,
            t.day_of_month,
            t.month_number,
            t.year,
            COUNT(*) AS rental_count,
            SUM(f.rental_cost) AS total_revenue
        FROM dw_fact_bike_rental f
        JOIN dw_dim_time t ON f.start_time_id = t.time_id
        JOIN dw_dim_station s ON f.start_station_id =
        s.station_id
        JOIN dw_dim_district d ON s.district_id = d.district_id
        JOIN dw_dim_city ci ON d.city_id = ci.city_id
        GROUP BY ci.city_name, t.day_of_month,
        t.month_number, t.year;


-- 1. Revenue share of each city within its country
        SELECT
            co.country_name,
            ci.city_name,
            SUM(f.rental_cost) AS city_revenue,
            ROUND(100 * SUM(f.rental_cost) /
        SUM(SUM(f.rental_cost)) OVER (PARTITION BY
        co.country_name), 2) AS revenue_share_percent
        FROM dw_fact_bike_rental f
        JOIN dw_dim_station s ON f.start_station_id =
        s.station_id
        JOIN dw_dim_district d ON s.district_id = d.district_id
        JOIN dw_dim_city ci ON d.city_id = ci.city_id
```

```
JOIN dw_dim_country co ON ci.country_id =
co.country_id
GROUP BY co.country_name, ci.city_name;
```

There is also a file called ‘sql_loader_proof.txt’ which stores results of executing load_all.bat.

3. Conclusion

We set up a data warehouse specifically for the analysis of bike rental data as part of this project. In addition to a central fact table that records rental transactions, the system was thoughtfully designed around a number of dimensions, including time, location, bike details, customers, memberships, payment methods, and weather. Flexible and effective analysis of different facets of bike rental operations, customer behavior, and environmental influences is made possible by the warehouse structure. We developed five query categories using ROLLUP, CUBE, PARTITION BY, VIEWS, and RANKING FUNCTIONS to improve analytical capabilities. These query categories produced informative aggregated reports, comparative analyses, and ranked results based on a variety of criteria.